POKÉMON
RED, BLUE AND YELLOW
CUSTOM MOTHERBOARD



PRODUCT

HTTPS://SHOP.GILTESA.COM/PRODUCT/POKEMON-RED-BLUE-AND-YELLOW-FOR-GAME-BOY-CUSTOM-MOTHERBOARD

PLEASE READ THROUGH THESE INSTRUCTIONS ENTIRELY BEFORE ATTEMPTING TO INSTALL.

WARNING: IF YOU ARE NOT COMFORTABLE WITH SOLDERING, OR PERFORMING ANY STEP IN THIS GUIDE, DO NOT PERFORM THE INSTALL YOURSELF. FIND SOMEONE WHO IS COMFORTABLE TO DO IT FOR YOU.

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DESCRIPTION

This is a **full-color printed custom motherboard** for the games **Pokémon Red, Pokémon Blue and Pokémon Yellow** for Game Boy.

You can use it to customize your original cartridge, whose battery has long since died or whose original PCB has become damaged or corroded.

Additionally, as you can see in the photos, this new circuit will give your Pokémon cartridge an incredible new look. Pair it with a transparent Game Boy or Game Boy Color shell to show off the artwork.

FEATURES

- New full-color printed motherboard for Pokémon Red, Blue and Yellow.
- Compatible with original Nintendo **MBC3** and **MBC5** donor cartridges.
- Replace the battery without soldering whenever it needs replacing.
- Compatible with GBFlash, allowing you to back up your save before the transplant and restore it afterwards.

INCLUDED

- 1x Full-color printed custom motherboard.
- 2x 10K 0603 resistors (1 required).
- 2x 33pF 0603 capacitors (1 required).
- 5x 10nF 0603 capacitors (4 required).
- 1x Battery holder.
- Custom vinyl sticker for the back with a Pokémon-style design.

RECOMMENDED / REQUIRED [NOT INCLUDED]

- An original Pokémon Red, Blue or Yellow donor cartridge matching the selected version.
- Transparent game cartridge for Game Boy or Game Boy Color. (1)
- New CR2032 button battery.
- 3.8mm gamebit screwdriver.
- Sandpaper or metal files (2)
- Tweezers.
- Tin soldering iron.
- Tin.
- Flux.
- Isopropyl alcohol.

(1) Remember to purchase the shell for the circuit. It's up to your discretion whether to use the one for Game Boy or Game Boy Color; both are compatible.

(2) Could be optional. In case it's necessary, it serves to reduce the side tabs that the circuit has. This depends on how faithful the shell is to the original; the tabs prevent the circuit from wobbling or shifting from side to side. If the circuit doesn't fit into the shell, you'll need to file the tabs slightly.

NOTES

This motherboard is designed for the MBC3A, MBC3B and MBC5 cartridge revisions used in most international releases. It is not compatible with the early Japanese MBC1 revisions. Please check the installation guide for full compatibility information before purchasing.

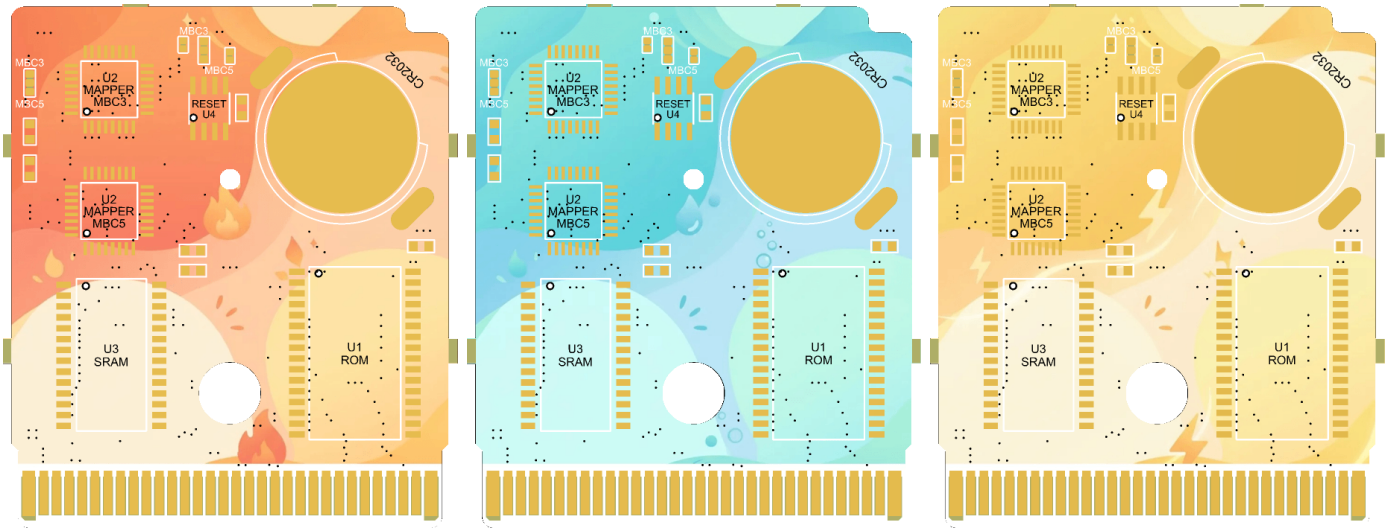
Although it is designed for use with Pokémon, it is also compatible with any other game whose motherboard has the identification number DMG-A02-01 or DMG-KFDN-10 and does not use an RTC.

BOARD DETAILS

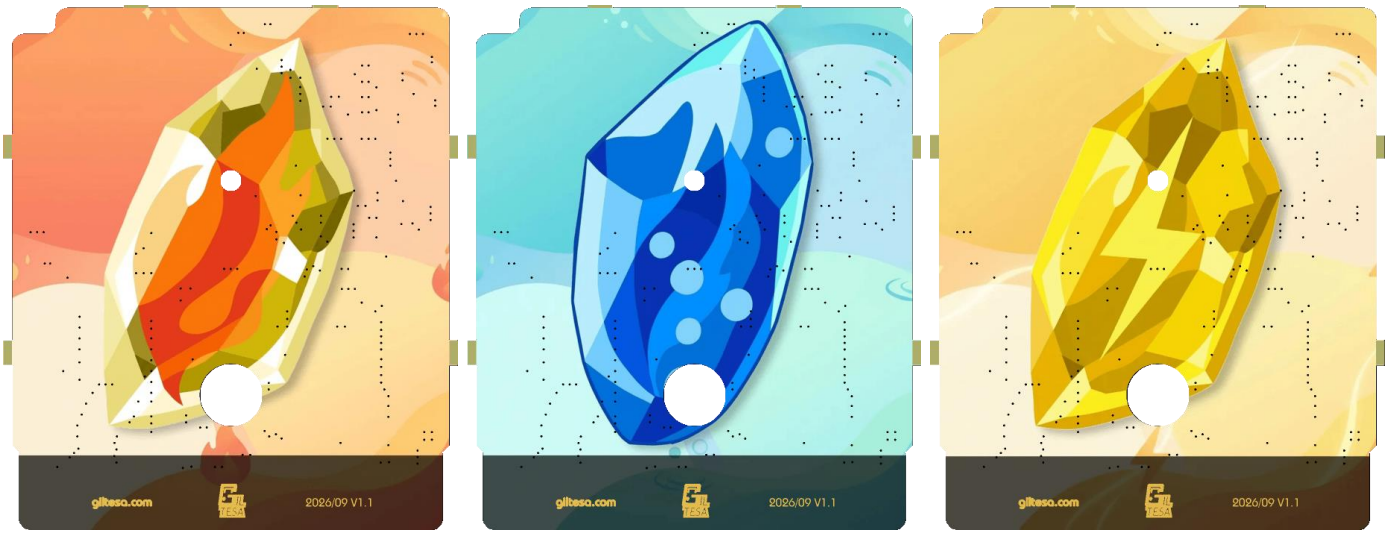
The board doesn't have any connectors or pads where wires need to be soldered like our other kits. However, this board doesn't come assembled; you will be the one to assemble the components supplied with the kit, as well as the 4 chips that make up the game, which you'll need to extract from your donor game.

All the steps are identical, regardless of the board color. In this guide, we will show you how to install it on Pokémon Blue.

Light-colored front design that makes it easier to solder the components onto the board and see them clearly once the cartridge is assembled.



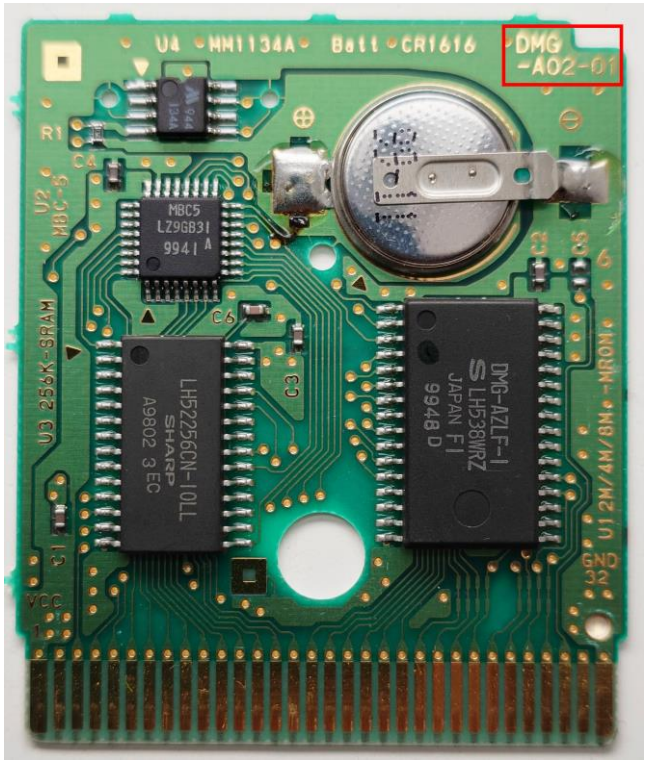
Fire, Water, and Electric Evolution Stone design on the back. A sticker is included that can optionally be used to give it a more recognizable look associated with the franchise.



INSTALLATION STEPS

Please, carefully read the following steps for a successful installation.

You can use any Pokémon donor with an **MBC3** or **MBC5** chip from American or European games, respectively. You can check whether yours is compatible by looking at the motherboard number. It should be either **DMG-A02-01** or **DMG-KFDN-10**. Earlier Japanese Pokémon games that came with an MBC1 are not compatible.





NOTE*: The board should also be compatible with any other Game Boy game that uses a **DMG-A02-01** or **DMG-KFDN-10** and does not use the RTC/clock function.*

PRE INSTALLATION STEPS

Before proceeding with the installation, **you need to retrieve the 4 chips that make up Pokémon game from your game circuit**. These chips must be transplanted to the new circuit.

This is where all the difficulty of the installation lies, as it is most convenient to use a hot air soldering station or a heating table to extract the chips. It is not recommended to use just a soldering iron, although perhaps in combination with a desoldering pump, it may be possible. The choice of tools is up to you.

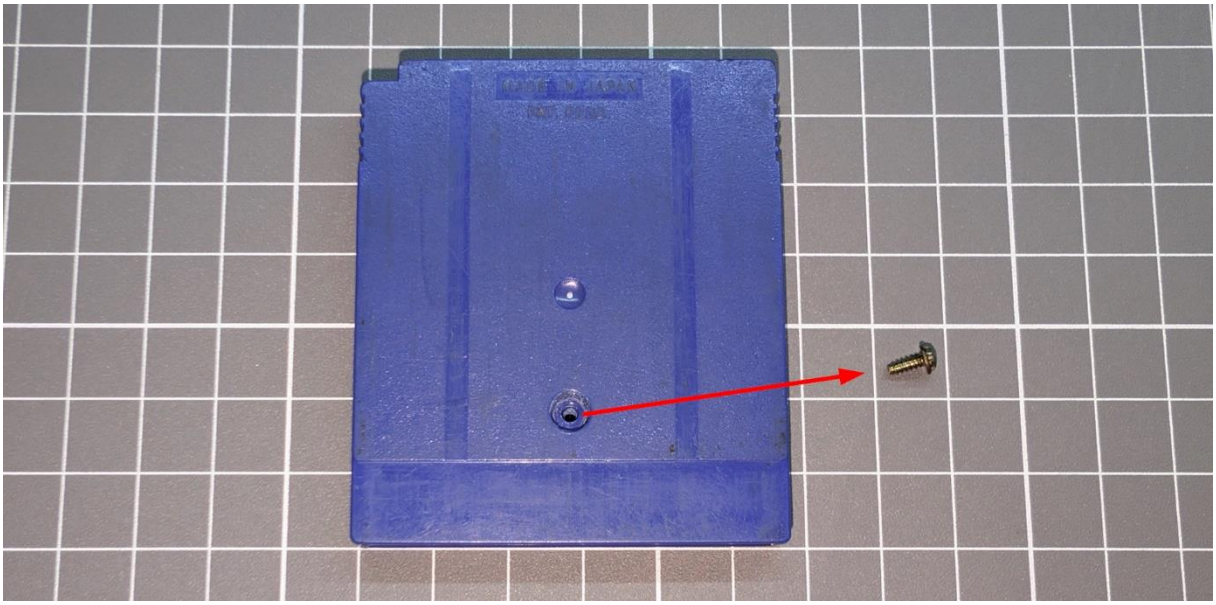
1. DISASSEMBLY THE CARD SHELL

First, you'll need to open the original game cartridge.

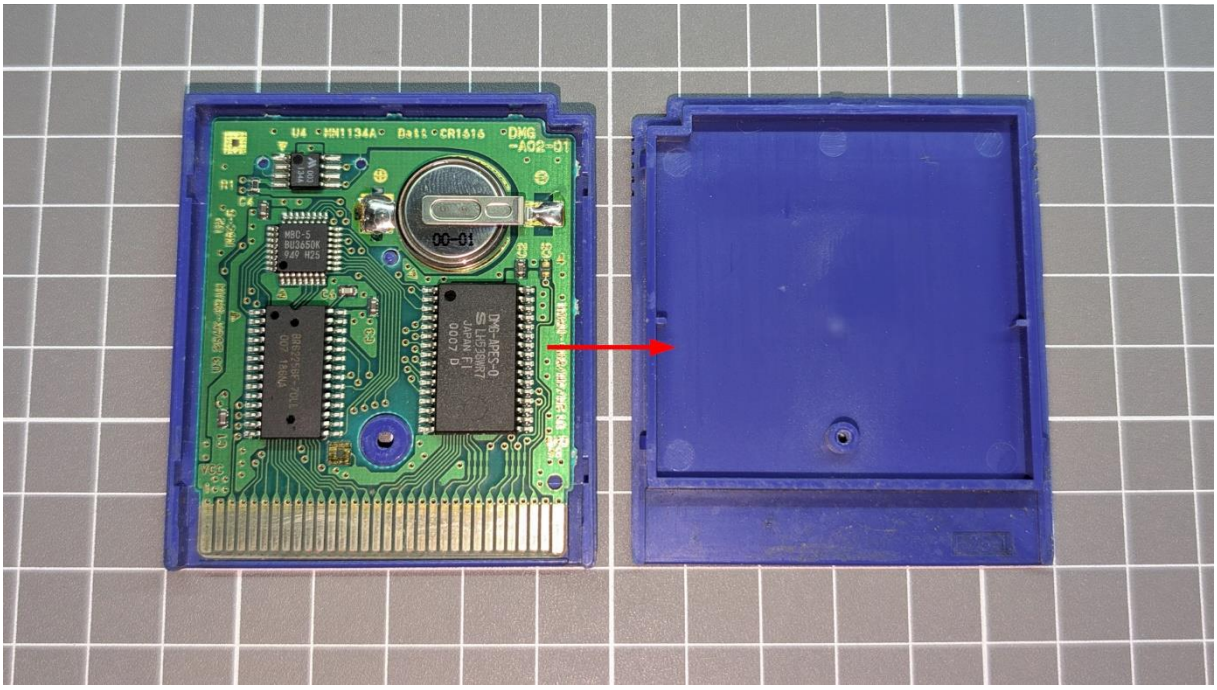


Gamebit screwdriver

The cartridge uses a **3.8mm gamebit screw**, this is the screw:



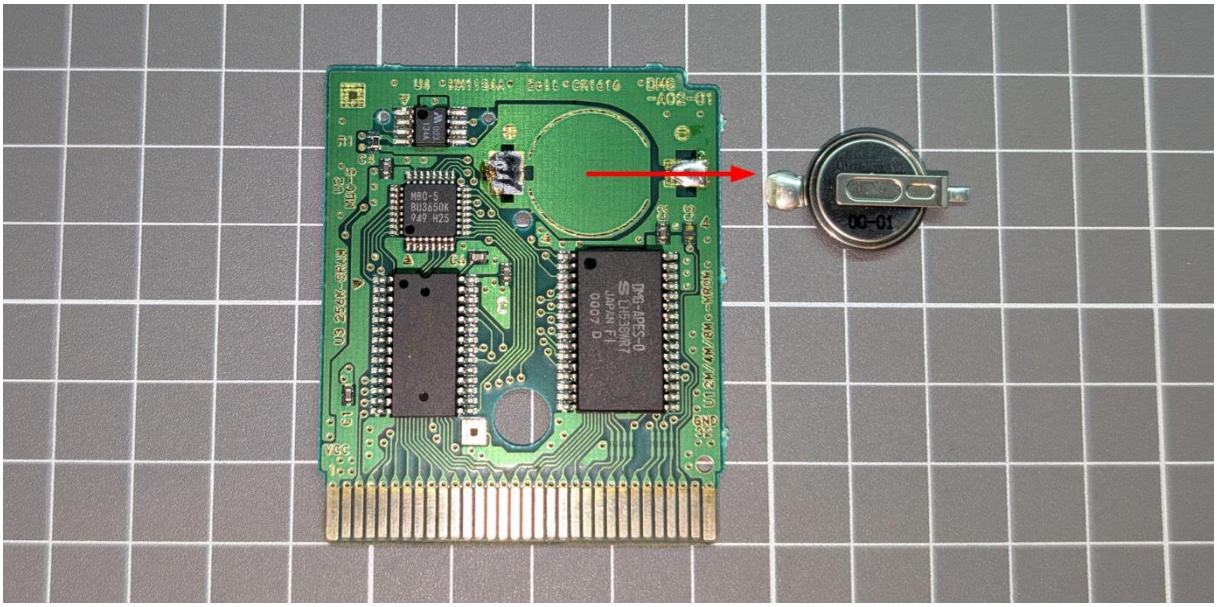
Once opened, you'll see the interior, just an electrical circuit where all the components are soldered.



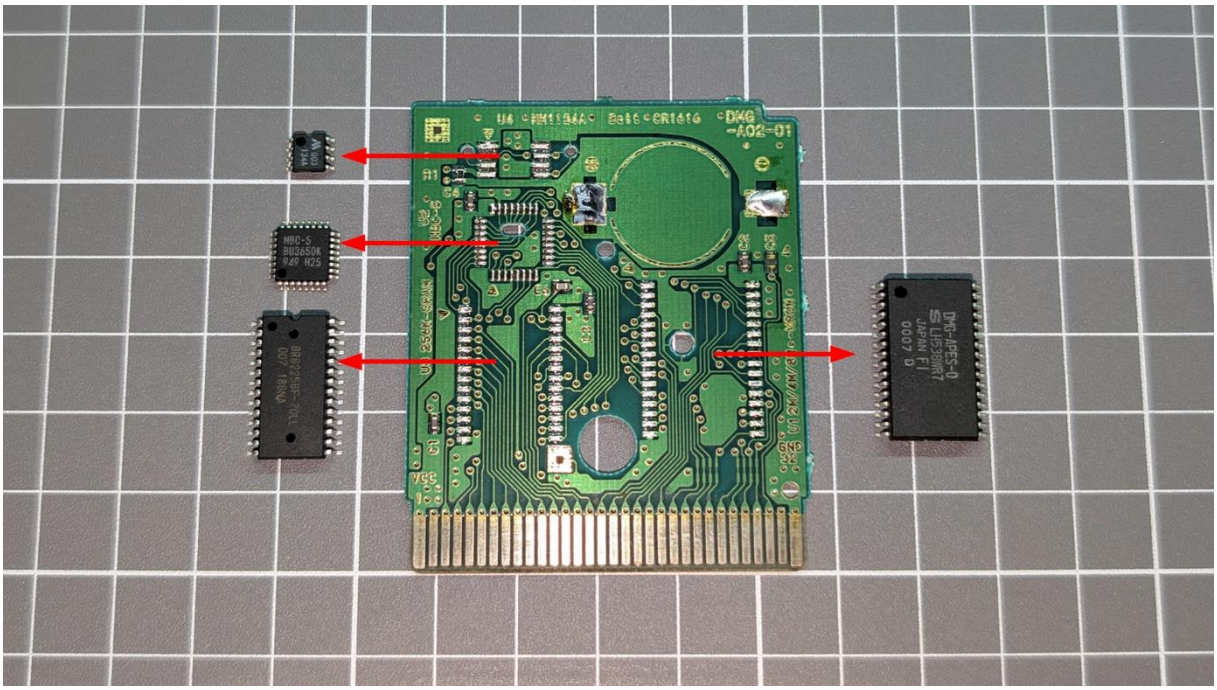
2. DISASSEMBLY THE CIRCUIT COMPONENTS

The first component to remove is the game's battery. No matter what type of tools you'll use to remove the rest of the components, this component must be removed manually with a soldering iron.

IMPORTANT NOTE: Removing the battery will erase any saved game data (if it hasn't already happened due to the battery running out). It is recommended to make a backup first if you don't want to lose your game progress.



Then comes the tricky part. It is recommended to use a hot plate or a hot air soldering station, in both cases heating the circuit from the underside so that the chips do not receive excessive heat. You can add a bit of flux to help the solder melt more quickly and thus free the chip to be removed with tweezers.



You don't need to remove the upper resistor or the three capacitors since new units are included in the kit.

INSTALLATION STEPS

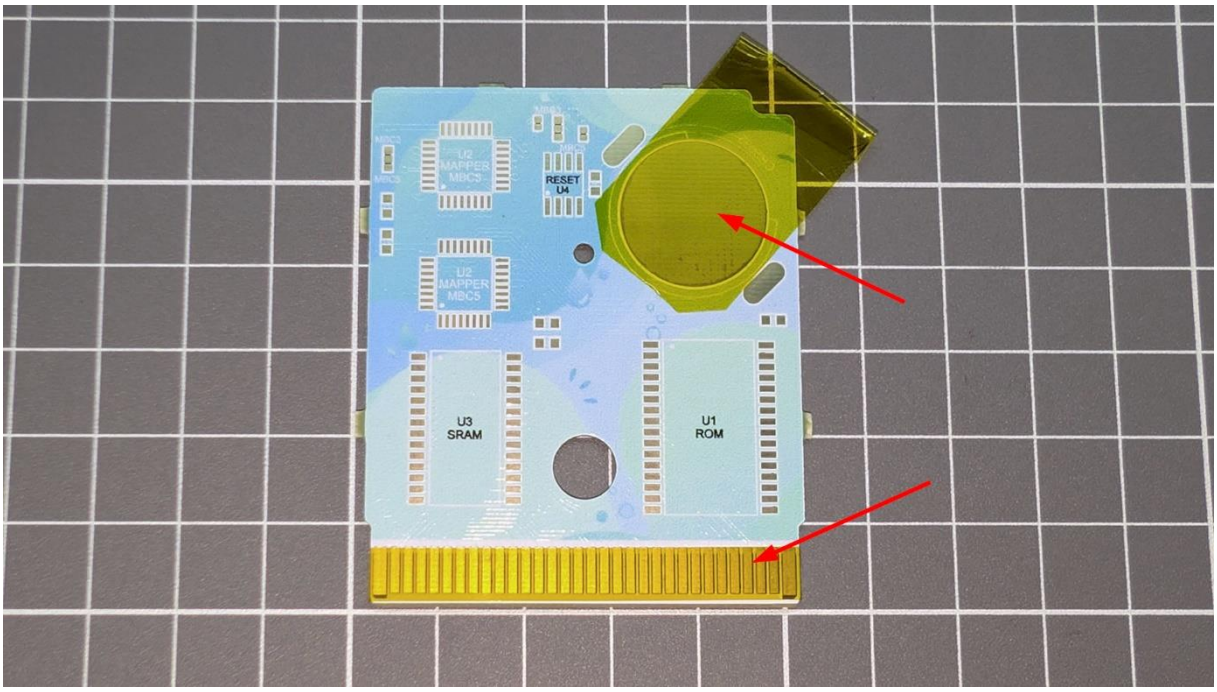
1. ISOLATION OF THE CIRCUIT

As you can see when opening the kit bag, the circuit already includes a kapton tape that protects the bottom where the pads are located, which will make electrical connection with the Game Boy game connector.

The negative terminal of the battery is also protected (in version 1.2 and later). Do not remove this until after soldering the battery holder.

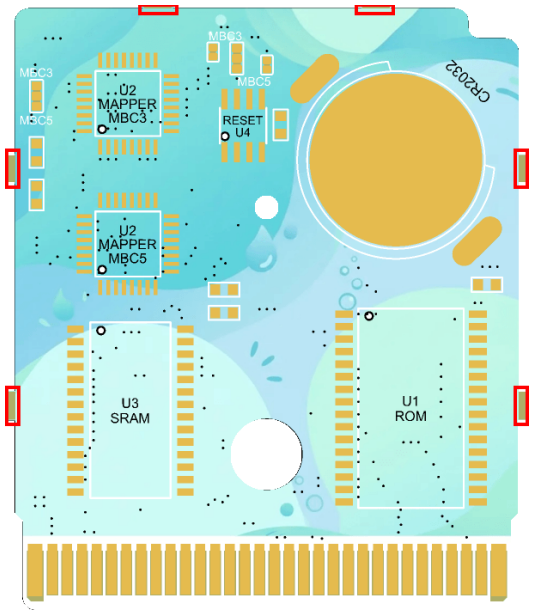
It's crucial to keep this kapton tape in place until you finish the installation completely. This prevents accidental solder from falling onto these pads and ruining the PCB.

If you have removed the kapton tape, put it back in place 😊

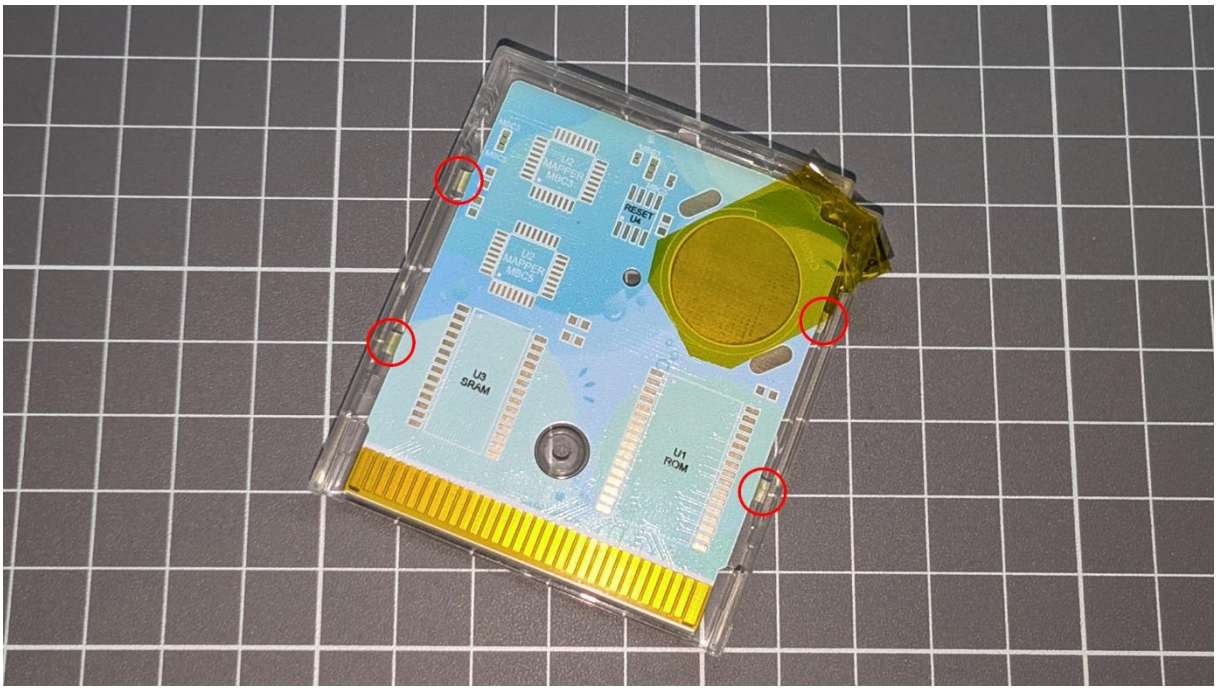


2. SANDING THE CENTERING TABS [OPTIONAL]

Since there's a wide variety of manufacturers producing card shells for Game Boy, not all of them have the same dimensions as the original shells. That's why the included circuit has side and top tabs to try to centre the circuit as much as possible within the shell.



Insert the circuit into the shell you want to use and check if it fits correctly. If you have trouble inserting it, you'll need to sand the tabs very slightly, first on one side and then on the other. Sand gradually and check if the circuit fits into place or not.



When you finish, clean the circuit with isopropyl alcohol.

TODO